

GRUZVER ROMARIO PHOCCO CÁCERES

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Summary

Robotics engineer and MSc student (Computer Science, PUCP) with hands-on experience in autonomous navigation, robot perception, and multi-robot systems for real-world deployment. Currently building a heterogeneous SAR robot team — quadruped, rover, and thermal UAV — for post-disaster victim detection under a PROCIENCIA national research grant. Core focus: ROS2 navigation stacks, thermal person detection, sim-to-real transfer, and reinforcement learning for autonomous systems. Invited to explore my work at github.com/Gruzver.

Education

Pontifical Catholic University of Peru (PUCP) **2026 – Present**

MSc in Computer Science | Specialization: Robotics & Reinforcement Learning

Pontifical Catholic University of Peru (PUCP) **August 2019 – August 2024**

Bachelor's Degree in Mechatronics Engineering

High-Performance School (Colegio de Alto Rendimiento) **2015 – 2018**

Secondary Education | Full government scholarship · IB Diploma (HL: Mathematics, Technology & Innovation)

Technical Skills

Robotics & Middleware:

- ROS2, Nav2, MoveIt 2, micro-ROS
- SLAM Toolbox, robot_localization (EKF)
- Ignition Gazebo, Isaac Sim (learning)

Perception & AI:

- Python (Advanced), C/C++ (Intermediate)
- PyTorch, TensorFlow, OpenCV
- Reinforcement Learning, Computer Vision

Hardware & Design:

- RealSense D435, RPLidar, ODrive
- Raspberry Pi, Jetson Nano/Orin
- DJI thermal cameras, Altium, KiCad

Dev Tools: Git/GitHub, Linux/Ubuntu, Docker, Bash **Languages:** Spanish (Native), English (B2), German (A1)

Research Experience

Research Assistant – Search & Rescue Robotics [github] **Jan. 2026 – Present**

Interdisciplinary Research Laboratory, PUCP

Lima, Peru

- Built an end-to-end ROS2 pipeline for real-time person detection from thermal drone video: YOLOv8x fine-tuned on ~5,000 custom thermal frames, multi-object tracking with persistent IDs, and Kalman filter GPS smoothing.
- Developed pixel-to-GPS geolocation using full drone telemetry (altitude, yaw/pitch/roll, FOV) and a live Leaflet.js operator dashboard for SAR teams.
- Planned and executed aerial data collection campaigns with DJI thermal drone; currently training multispectral detection model for post-disaster scenarios.

Research Assistant – Social Robotics (AMII) [github] **Sep. 2024 – Mar. 2026**

Interdisciplinary Research Laboratory, PUCP

Lima, Peru

- Designed and deployed the full locomotion system for AMII, a differential-drive social robot: mechanical design, electrical integration, and ROS2 Nav2 stack (AMCL, SLAM Toolbox, EKF, RealSense D435 obstacle avoidance).
- Developed agentic NLP interaction stack enabling autonomous engagement with people in real public environments using LLM-based dialogue.
- Integrated navigation, hardware drivers, and natural language control into a unified deployable system; coordinated live demos at public events.

Work Experience

Lab Supervisor / Teaching Assistant

August 2024 – Present

Engineering Department, PUCP

Lima, Peru

- Teaching assistant for Robotics & AI and Mechatronics Design Project courses (Mechatronics Engineering program).
- Developed educational platforms: omnidirectional mobile robot and 4-DOF robotic arm with MoveIt 2 and micro-ROS on Raspberry Pi Pico W.

Design and Product Development Intern

March 2024 – September 2024

FABRICUM PUCP

Lima, Peru

- Designed and built a 3D printer with infinite Z-axis printing capabilities; led multidisciplinary team through full hardware and software development cycle.
- Applied CAD modeling (Fusion 360, SolidWorks) and 3D prototyping for iterative mechanical design and validation.

Project Intern – Interdisciplinary Research Laboratory

October 2023 – March 2024

Faculty of Science and Engineering, PUCP

Lima, Peru

- Developed TS, a functional Add-In for Autodesk Inventor enabling rapid prototyping workflows; software development with SCRUM methodology.

Innovation Intern – Strategy and Development

January 2023 – June 2023

Ferreycorp

Lima, Peru

- Coordinated Sitech's application to ProInnovate open innovation contest (non-reimbursable funding up to \$554,400 USD); process mapping and startup partnership management.

Key Projects

Chaska Rover | ROS2, MoveIt 2, Pinocchio IK, 3D LiDAR | For URC/ERC Competitions | [github](#) Oct. 2025 – Present

- Led design and deployment of a 6-DOF robotic arm: mechanical design, Gazebo simulation, MoveIt 2 motion planning, and AI object detection on hardware.
- Developed multi-mode locomotion (swerve, differential, Ackermann) with velocity-based IK, real-time ROS2 control, and PS5 DualSense teleoperation.

Kit Manipulador | ROS2, MoveIt 2, micro-ROS, Gazebo, Raspberry Pi Pico W | [github](#)

2024

- 4-DOF robotic arm with full ROS2 + MoveIt 2 motion planning pipeline, Gazebo simulation, and real hardware integration via micro-ROS on Raspberry Pi Pico W.

PUCP Intelligent Robotics Group | Reinforcement Learning, AWS DeepRacer, ROS2

Mar. 2024 – Present

- Autonomous navigation training with RL on AWS DeepRacer; developed ROS2-controlled arm for autonomous chess playing with CV-based piece detection.

Achievements

RoboCup ARM Challenge | Finalist

2026

- Selected as finalist to compete in the RoboCup 2026 World Championship (Incheon, South Korea), developing an autonomous robot arm pipeline with RGB-D point cloud perception, object classification, and pick-and-place manipulation using MATLAB/Simulink and a UR5e.

AI for Industry Challenge (Intrinsic) | Top Qualifier | [github](#)

2026

- Developed a vision-based fiber optic connector insertion policy for a UR5e robot, achieving ~224/300 in the qualification phase. Pipeline: YOLOv8-Pose keypoint detection, solvePnP 6-DOF pose estimation with multi-camera fusion, and impedance-controlled spiral insertion with F/T snap detection.

Aspire Leaders Program | Cohort 2, Harvard University

2026

- Selective leadership development program at Harvard University for emerging global leaders, focusing on professional growth, systems thinking, and social impact.

Desafiata PUCP | Innovation Contest Winner

2022

- Led team of 4 to win PUCP's health innovation contest with Wayky, an AI virtual assistant for university student wellbeing.

Innova and Business Fest PUCP | 2nd and 3rd Place

2023

- Presented AR applications for Interbank (virtual navigation, 2nd place) and Cálidda (pipe/valve detection, 3rd place).